INCH-POUND
MS27404C
19 January 2017
SUPERSEDING
MS27404B
w/AMENDMENT 2
20 November 2013

DETAIL SPECIFICATION SHEET

ADAPTER, STRAIGHT, REUSABLE, TUBE TO HOSE, LOW PRESSURE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38726.

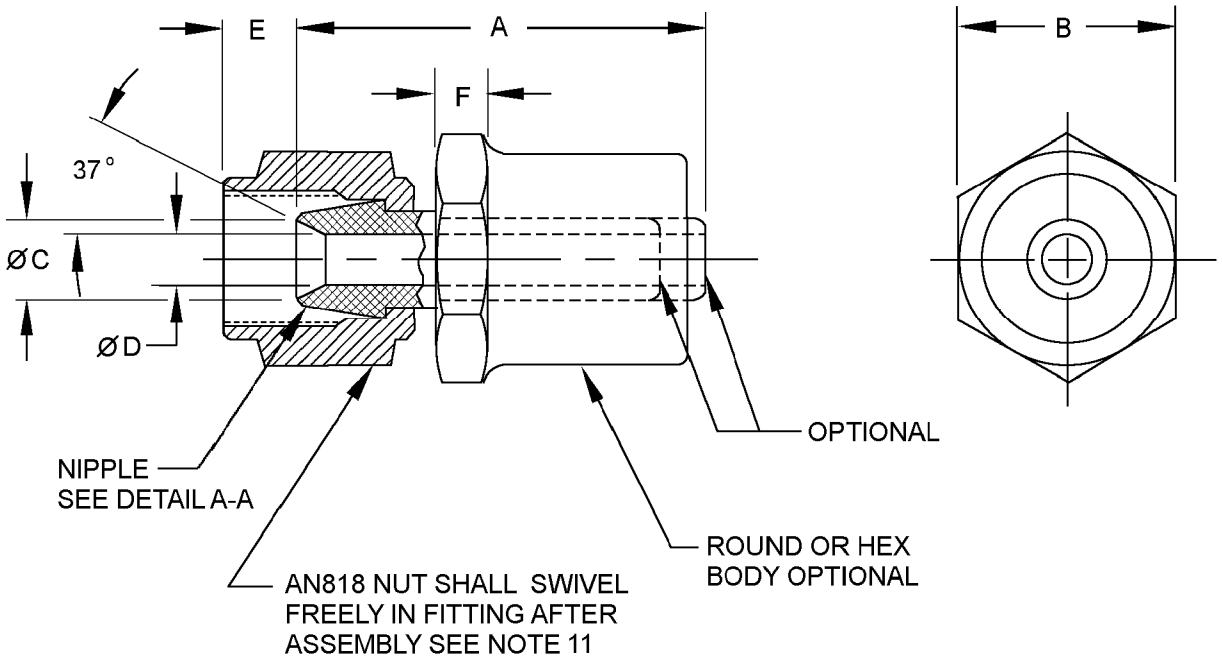
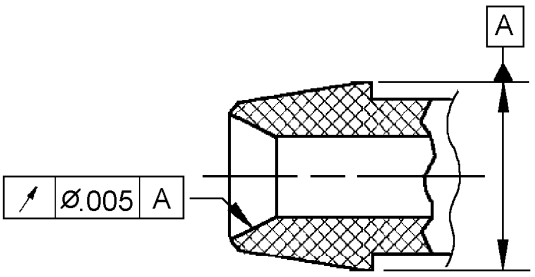


FIGURE 1. Adapter dimensions and configuration.

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DETAIL A-A

Dash number	Tube OD inches (mm)	A max inches (mm)	B max (see note 7) inches (mm)	C (see note 8) ±.010 (±0.25) inches (mm)	D min inches (mm)
-2	.125 (3.18)	1.156 (29.36)	.500 (12.70)	.195 (4.95)	.052 (1.32)
-3	.188 (4.78)	1.219 (30.96)	.563 (14.30)	.245 (6.22)	.109 (2.77)
-4	.250 (6.35)	1.250 (31.75)	.625 (15.88)	.295 (7.49)	.156 (3.96)
-6	.375 (9.53)	1.625 (41.28)	.813 (20.65)	.435 (11.05)	.281 (7.14)
-8	.500 (12.70)	1.750 (44.45)	1.000 (25.40)	.570 (14.48)	.375 (9.53)
-10	.625 (15.88)	1.875 (47.63)	1.125 (28.58)	.680 (17.27)	.453 (11.51)

Dash number	E +.016/-.000 (+0.41/-0.00) inches (mm)	F min inches (mm)	Nut AN818
-2	.328 (8.33)	.188 (4.78)	AN818-2
-3	.313 (7.95)	.188 (4.78)	AN818-3
-4	.344 (8.74)	.188 (4.78)	AN818-4
-6	.375 (9.53)	.250 (6.35)	AN818-6
-8	.422 (10.72)	.250 (6.35)	AN818-8
-10	.500 (12.70)	.313 (7.95)	AN818-10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Dimensioning and tolerancing in accordance with ASME Y14.5.
4. Unless otherwise specified, tolerances are ±.005 inch (0.25 mm), angular dimensions ±0°30’.
5. Break sharp edges and remove all hanging burrs and slivers.
6. Surface roughness shall be 125 micro inches Ra in accordance with ASME B46.1.
7. Dimension B shall fit standard wrench opening.
8. Dimension C applies to machined parts only.
9. When a flared tubing end is used instead of a machined part, the flare shall be in accordance with SAE-AS4330.
10. For design features purposes, this standard takes precedence over documents referenced herein.
11. Reference AN818 for complete PIN. Internal threads shall be in accordance with AN818 nut.

FIGURE 1. Adapter dimensions and configuration - Continued.

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REQUIREMENTS:

Adapter shall be as specified on figure 1.

Fitting shall withstand all tests specified in MIL-DTL-38726 when assembled to hose in accordance with MIL-DTL-5593.

Materials shall be in accordance with table I.

TABLE I. Adapter material finish and identification codes.

Designator	Material
D 1/	Aluminum alloy 2014-T6 or 2024-T6 or -T851
J	Corrosion resistant steel (CRES), type 304
K	Corrosion resistant steel (CRES), type 316
P	Chrome-molybdenum steel 4130
S	Steel 4140
T 2/	Titanium
W	Aluminum alloy 7075-T73

1/ Inactive for new design, for new design use aluminum alloy 7075 to improve stress corrosion resistance and tensile strength.
2/ Titanium shall not be used in oxygen systems.

Finish. Finishes shall be as specified in table II. All platings shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion of the base metal after 96 hours of salt spray. Fluid passages, other openings, and internal threads shall not be subject to the plating thickness requirement and may have bare areas, provided they are protected with a light film of oil.

Adapter color coding. Adapter color coding shall be in accordance with MIL-DTL-38726.

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TABLE II. Material finish identification codes.

Plating finish designator	Material	Plating Finish
Blank	Steel 4130 or 4140	Cadmium 1/
Blank	Aluminum	Anodized
AN		Anodized then apply NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706
H	Steel 4130 or 4140	Zinc aluminum in accordance with ASTM F1136/F1136M, grade 3, NC.
J	Steel 4130 or 4140	Zinc-nickel 2/
P	Steel 4130 or 4140	Zinc phosphate in accordance with MIL-DTL-16232, type Z, Class 4
R	Steel 4130 or 4140	Zinc plating in accordance with ASTM B633 types III & V, Fe/Zn 8
Blank	CRES	No additional finish, passivated
Blank	Titanium	Anodize in accordance with SAE-AMS2488, type 2
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486
Z	Steel 4130 or 4140	Zinc plating in accordance with ASTM B695, type I, class 8 3/
ZC	Steel	Zinc may be any zinc plating from PIN codes J, R, and Z with a colored chromate coating 3/

1/ Cadmium shall not be used in oxygen or potable water systems.
2/ Zinc nickel shall be in accordance with SAE-AMS2417, type 2, grade B, except the zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum coating thickness.
3/ Not for use in aircraft. Requires approval from the Program Officer for all applications.

Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using the brush plating process below. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS2451/5.

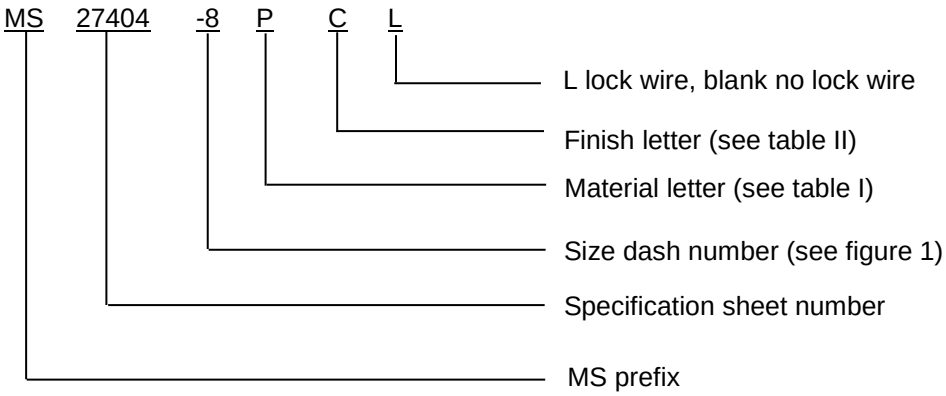
b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS2451/9.

c. Brush plating of NAVAIR TCP shall be in accordance with MIL-DTL-81706, type II, class 1A, material form 1 through 6, application method B. Example of a PIN: M817062A6B.

NOTE. To the users of this document it is recommended that the use of steel parts with cadmium plating be used only when other materials and finishes specified in the document cannot meet performance requirements.

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Part or Identifying Number (PIN): The PIN consists of the letter “MS” the specification sheet number, a dash number for tube and port size, a letter designator for material type, material finish letter, and a blank (no lockwire) or L for lockwire. Unassigned PIN’s shall not be used.



PIN examples:

MS27404-8S is for a straight adapter, hose to tube, .500 inch (12.70 mm), steel (4140) with cadmium plate.

MS27404-8PCL is for a straight adapter, hose to tube, .500 inch (12.70 mm), steel (4130) with cadmium and a nut drilled for lockwire hole.

MS27404-8WL is for a straight adapter, hose to tube, .500 inch (12.70 mm), aluminum alloy 7075-T73, AN818 nut drilled for lockwire hole.

NOTE: Cadmium plating is not recommended. Carbon steel material with cadmium plating shall only be used when other materials and finishes specified in this document cannot meet performance requirements.

Marking: Part shall be permanently marked with the MS PIN, and include the manufacturer’s CAGE, name, or trademark.

Referenced documents shall be in effect on the date of invitation for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-38726, this document references the following:

AN818	ASTM B117	SAE-AMS2417
MIL-DTL-5593	ASTM B633	SAE-AMS2486
MIL-DTL-16232	ASTM B695	SAE-AMS2488
MIL-DTL-81706	ASTM F1136/F1136M	SAE-AS4330
ASME B46.1	SAE-AMS2451/5	
ASME Y14.5	SAE-AMS2451/9	

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CONCLUDING MATERIAL

Custodians:
Army - MI
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2017-027)

Review activities:
Army - AV
Navy - SA
Air Force - 71

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